PS: 2 Name:

1. Have you installed R on your computer? Do you have access to the command line via a shell? This will likely be Terminal on a Mac (which is built-in) or git bash on Windows.

R on the File System

Assume you start a new R session with your working directory at `~`. Which R functions would you call to do each of the following?

```
~
├── Project-1
│   ├── raw-data-1.txt
│   ├── processed-data-1.csv
│   └── data-processing-1.r
├── Project-2
│   ├── raw-data-2.txt
│   ├── processed-data-2.csv
│   └── data-processing-2.r
```

2. Change your working directory to be `Project-1`?

3. Read the data in `raw-data-1.txt` and save it to the R object named `raw_data_1`.

4. Without changing your working directory, read the data in `raw-data-2.txt` and save it to the R object named `raw_data_2`.

5. You may have noticed that when creating R objects, I avoid certain types of names. What characters are permissable when naming new objects?[1]

[1] You can research the answer to this question online.

Atomic Vectors

If you recorded data on peoples' answers to the following questions and read each one into R as a vector, what data type could be used to store them?

6. How much do you spend on average per week, in dollars and cents, on caffeinated beverages?

7. Did you vote in the last US Election?

8. How many US states have you visited?

9. Which US states have you visited?

While each of the following vectors *could* be created by simply using `c()`, instead write a command that can generate each one more efficiently using the functions from class. Read more aboutthem by pulling up their help files with `?`.

10. `_________________________`

```
 [1]  2  4  6  8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48
```

11. `_________________________`

```
 [1] 10  9  8  7  6  5  4  3  2  1
```

12. `_________________________`

```
[1] "A" "A" "B" "B" "B" "C" "C" "C" "C"
```

13. `_________________________`

```
 [1]  0  0  0  5  5  5 10 10 10 15 15 15
```

14. What single command can be run on each of these vectors

```
x <- c("A", "B", "C")
x <- seq(1, 5, 1)
x <- c(TRUE, FALSE)
```

and produce this output?

```
[1] "A" "B" "B" "C" "C" "C"
```

```
 [1] 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5
```

```
[1]  TRUE FALSE FALSE
```

Working with Vectors

Coming soon.